Chapter 10

Parametric Bootstrap

What is known as a *parametric bootstrap* is a widely applicable method for assessing uncertainty in parameter estimates, often in the form of interval estimation, although the formation of prediction regions or intervals is also a clear area of application. We will present the parametric bootstrap in the case of a scalar parameter θ, although it appears that the same essential ideas could be easily extended to confidence regions for vector-valued parameters. In particular, since the parametric bootstrap is a version of simulation-based inference, using a model with $\boldsymbol{\theta} \equiv (\theta_1, \boldsymbol{\theta}_2)$ and simulating in the manner to be described below results in an assessment of the marginal distribution of an estimator $\hat{\theta}_1$. Parametric bootstrap may certainly be applied to problems in which we are able to locate simultaneous maximum likelihood estimates, but more often it sees use with estimators of alternative forms such as those we have discussed in the past several chapters.

10.1 Basic Bootstrap Estimators

To set the basic notation for this section, let $Y_1, \ldots, Y_n$ be random variables that follow a model giving joint density or mass function $f(y_1, \ldots, y_n|\theta)$. Suppose that an estimator of θ, $\hat{\theta}$, is available by some means (maximum likelihood, least squares, quasi-likelihood, an estimating function, or composite likelihood). We assume that $\hat{\theta}$ is a function of the observations, $\hat{\theta} \equiv \hat{\theta}(\boldsymbol{y})$, whether or not that function can be expressed in closed form. Substitution of $\hat{\theta}(\boldsymbol{y})$ for θ in the model gives the *fitted model* with density or mass function $f(y_1, \ldots, y_n|\hat{\theta}(\boldsymbol{y}))$ and distribution function $F(y_1, \ldots, y_n|\hat{\theta}(\boldsymbol{y}))$. In these expressions it must be understood that the arguments $y_1, \ldots, y_n$ are arbitrary, while $\boldsymbol{y}$ in $\hat{\theta}(\boldsymbol{y})$ is the observed data from a particular study. The basic simulation process is to generate observations $\boldsymbol{y}^* = (y_1^*, \ldots, y_n^*)$ from the fitted model and then calculate the estimator of θ from these simulated values, which we will denote $\hat{\theta}(\boldsymbol{y}^*)$. If this process is repeated M times, we obtain the bootstrap estimates $\{\hat{\theta}(\boldsymbol{y}_m^*) : \ m = 1, \ldots, M\}$. The underlying idea is that the distribution of a function of the true parameter and the actual estimate, $h(\theta, \hat{\theta}(\boldsymbol{y}))$, can be approximated by the empirical distribution of that same function of $\hat{\theta}(\boldsymbol{y})$ and the bootstrap estimates $\hat{\theta}(\boldsymbol{y}_m^*)$.

What we have is essentially a Monte Carlo procedure with a twist. Rather than computing a Monte Carlo approximation to the expected value of a simple function of $\hat{\theta}$, we are attempting to obtain a Monte Carlo approximation to the expected value of a function of $\hat{\theta}$ and the true parameter value θ. Because the true parameter is unknown, we use the actual estimate $\hat{\theta}(\boldsymbol{y})$ as the true parameter in simulation. The reason for using a function of estimator and parameter $h(\theta, \hat{\theta}(\boldsymbol{y}))$ is to mitigate the effect of the true parameter on the sampling distribution of the estimator. A simple example will help make

this clear. Consider estimating the mean of a normal distribution μ based on a data set $\boldsymbol{y}$ assumed to be a realization of a one-sample normal model. The estimator, no matter how you approach the problem, is the sample mean $\hat{\mu}(\boldsymbol{Y}) = \bar{Y}$. The sampling distribution of $\bar{Y}$ depends on the parameter μ, but the distribution of $h(\mu, \bar{Y}) = \bar{Y} - \mu$ does not. Thus, the empirical distribution of $\{h(\bar{y}, \bar{y^*}_m) = (\bar{y^*}_m - \bar{y});\ m = 1, \ldots, M\}$ will not depend on the actual estimate $\bar{y}$ and can be taken as an approximation to the distribution of $h(\mu, \bar{Y})$. Now, it is not always possible to determine a function $h(\theta, \hat{\theta}(\boldsymbol{Y}))$ that is a pivotal quantity, although that is the ideal. The concept is that the use of this comparison or discrepancy function can help reduce, if not eliminate, the dependence of the sampling distribution of $\hat{\theta}(\boldsymbol{Y})$ on the parameter θ.

The point of the previous paragraph is essential but is also easily forgotten or ignored because Monte Carlo estimators using one of the most common discrepancy functions, the simple difference, often appear to simply use the empirical distribution of bootstrap estimates at the true parameter value $\hat{\theta}(\boldsymbol{Y})$ to approximate the sampling distribution of the estimator $\hat{\theta}(\boldsymbol{Y})$ at the actual true parameter θ. Although it increases our notational burden a bit, a practical mechanism to avoid making errors is to only bootstrap quantities that have asymptotic distributions. In our context this means that disrepancy functions chosen for use in applications should always correspond to appropriately centered and scaled versions of estimators for which a limit distribution is known to exist. This serves as a very effective way to avoid committing potentially serious mistakes. To simplify notation at this point we will write $\hat{\theta}_n$ rather than $\hat{\theta}(\boldsymbol{y})$ and $\theta^*_{n,m}$ rather than $\hat{\theta}(\boldsymbol{y}^*_m)$. Consider a maximum likelihood estimator $\hat{\theta}(\boldsymbol{y})$ that satisfies typical regularity conditions. Then we know that $n^{1/2}(\hat{\theta}_n - \theta)$ has a limit distribution and we might

take

$$h(\theta, \hat{\theta}_n) = n^{1/2}(\hat{\theta}_n - \theta).$$

Consider approximating the expected value of this discrepancy function,

$$\begin{aligned} E_M\{h(\theta, \hat{\theta})\} &= E_M\{n^{1/2}(\hat{\theta}_n - \theta)\} \\ &= \frac{1}{M}\sum_{m=1}^{M} n^{1/2}(\theta_n^* - \hat{\theta}_n)_m \\ &= n^{1/2}(\bar{\theta^*}_n - \hat{\theta}_n). \end{aligned} \tag{10.1}$$

An approximation to the variance of $h(\theta, \hat{\theta}_n)$ is,

$$\begin{aligned} V_M\{h(\theta, \hat{\theta}_n)\} &= V_M\{n^{1/2}(\hat{\theta}_n - \theta)\} \\ &= \frac{n}{M}\sum_{m=1}^{M}\left[(\theta_n^* - \hat{\theta}_n) - (\bar{\theta^*}_n - \hat{\theta}_n)\right]_m^2 \\ &= \frac{n}{M}\sum_{m=1}^{M}(\theta_{n,m}^* - \bar{\theta^*}_n)^2. \end{aligned} \tag{10.2}$$

Now an approximation to $\text{bias}(\hat{\theta}_n) = E\{\hat{\theta}_n - \theta)\}$ is easily obtained from (10.1) and an approximation to $var(\hat{\theta}_n - \theta) = var(\hat{\theta}_n)$, may be obtained from (10.2).

The fact that we are not using the empirical distribution of bootstrap estimates as a *direct* approximation to the sampling distribution of $\hat{\theta}_n$ may become more clear if we consider using the empirical distribution function of $\{h(\hat{\theta}_n, \theta_{n,m}^*;\ m = 1, \ldots, M\}$ to approximate the distribution function of $h(\theta, \hat{\theta}_n)$. In this case,

$$\begin{aligned} G_M(u) &= \frac{1}{M}\sum_{m=1}^{M} I[h(\hat{\theta}_n, \theta_n^*)_m \le u] \\ &= \frac{1}{M}\sum_{m=1}^{M} I[n^{1/2}(\theta_{n,m}^* - \hat{\theta}_n) \le u] \\ &= \frac{1}{M}\sum_{m=1}^{M} I[\theta_{n,m}^* \le \hat{\theta}_n + u/n^{1/2}]. \end{aligned} \tag{10.3}$$

In (10.3) $I(A)$ is the indicator function that assumes a value of 1 if A is true and a value of 0 otherwise. It should now be clear that in parametric bootstrap methods the empirical distribution of bootstrap estimates is not being used to approximate the distribution of $\hat{\theta}_n$. In fact, the sampling distribution of $\hat{\theta}_n$ is not being approximated at all by these methods. It is the sampling distribution of $h(\theta, \hat{\theta}_n)$ that is the target of approximation.

In order to form bootstrap interval estimates of θ we will need to approximate specific quantiles of the distribution of $h(\theta, \hat{\theta}_n)$. To do so we use the general result that, if $X_1, \ldots, X_N$ are independently distributed with distribution function F_X, and if $X_{[1]}, \ldots, X_{[N]}$ denote the ordered values, then

$$E\{X_{[k]}\} \approx F^{-1}\left(\frac{k}{N+1}\right),$$

leading to the estimate of the qth quantile of F_X, which is $x_q = F_X^{-1}(q)$ as,

$$x_q = X_{[(N+1)q]}.$$

Thus, the estimated value of the qth quantile of $h(\theta, \hat{\theta}_n)$ is the $(M+1)q^{th}$ largest value of $\{h(\hat{\theta}_n, \theta^*_{n,m}) : m = 1, \ldots, M\}$, assuming that $(M+1)q$ is an integer, which we can always arrange, or we can use the largest integer less than or equal to $(M+1)q$ in its place.

10.2 Bootstrap Confidence Intervals

In this section we will consider three types of bootstrap confidence intervals the most important being what are called basic bootstrap intervals. Basic intervals developed using different discrepancy functions $h(\theta, \hat{\theta}_n)$ result in what are sometimes considered different types of intervals, but it is only the particular form chosen for this function that actually changes.

10.2.1 Normal Approximation Intervals

Consider a situation in which we would be willing to form intervals based on asymptotic normality of $\hat{\theta}_n$, but the variance or standard error of the limit distribution is unavailable or extremely difficult to compute. If $\hat{\theta}_n$ is AN with mean $\theta + B(\hat{\theta}_n)$ and variance $V(\hat{\theta}_n)$ then an approximate normal interval can be derived in the usual way from,

$$Pr\{L_\alpha \leq \theta \leq U_\alpha\} = 1 - \alpha,$$

which leads to

$$\begin{aligned} L_\alpha &= \hat{\theta}_n - B(\hat{\theta}_n) - V^{1/2}(\hat{\theta}_n) z_{1-\alpha/2} \\ U_\alpha &= \hat{\theta}_n - B(\hat{\theta}_n) + V^{1/2}(\hat{\theta}_n) z_{1-\alpha/2}, \end{aligned} \tag{10.4}$$

where $z_{1-\alpha/2}$ is the $1 - \alpha/2$ quantile of the standard normal distribution. To see this, note that it is the quantity

$$\frac{\hat{\theta}_n - B(\hat{\theta}_n) - \theta}{V^{1/2}(\hat{\theta}_n)},$$

that is approximately $N(0, 1)$. The derivation then proceeds exactly as in elementary statistics courses using $\hat{\theta}_n - B(\hat{\theta}_n)$ in place of an unbiased estimator of θ.

To use the interval (10.4) in practice requires only estimation of $B(\hat{\theta}_n)$ as given in (10.1) and $V(\hat{\theta}_n)$ as given in (10.2). A bootstrap normal approximation interval for θ is then,

$$\left(\hat{\theta}_n - B_M(\hat{\theta}_n) - V_M^{1/2}(\hat{\theta}_n) z_{1-\alpha/2},\ \ \hat{\theta}_n - B_M(\hat{\theta}_n) + V_M^{1/2}(\hat{\theta}_n) z_{1-\alpha/2}\right). \tag{10.5}$$

10.2.2 Basic Bootstrap Intervals

Now suppose we are in a situation in which we are reluctant to make direct use of an asymptotic result to compute intervals. We can derive an interval

with equal probability in each tail, that is, equal left-tail and right-tail errors by finding L_α and U_α such that

$$\begin{aligned} Pr\{\theta \leq L_\alpha\} &= \alpha/2 \\ Pr\{\theta \leq U_\alpha\} &= 1-\alpha/2. \end{aligned} \tag{10.6}$$

If $h(\theta, \hat{\theta}_n)$ does not reverse inequalities then

$$\begin{aligned} Pr\{h(\theta, \hat{\theta}_n) \leq h(L_\alpha, \hat{\theta}_n)\} &= \alpha/2 \\ Pr\{h(\theta, \hat{\theta}_n) \leq h(U, \hat{\theta}_n)\} &= 1-\alpha/2. \end{aligned} \tag{10.7}$$

Then $h(L_\alpha, \hat{\theta}_n)$ is the $\alpha/2$ quantile and $h(U_\alpha, \hat{\theta}_n)$ is the $1-\alpha/2$ quantile of the distribution of $h(\theta, \hat{\theta}_n)$. Let these be dentoed as $v_{\alpha/2}$ and $v_{1-\alpha/2}$, respectively. If $h(\theta, \hat{\theta}_n)$ does reverse inequalities, then

$$\begin{aligned} Pr\{h(\theta, \hat{\theta}_n) \leq h(L_\alpha, \hat{\theta}_n)\} &= 1-\alpha/2 \\ Pr\{h(\theta, \hat{\theta}_n) \leq h(U, \hat{\theta}_n)\} &= \alpha/2. \end{aligned} \tag{10.8}$$

In this case, $h(L_\alpha, \hat{\theta}_n) = v_{1-\alpha/2}$ is the $1-\alpha/2$ quantile and $h(U_\alpha, \hat{\theta}_n) = v_{\alpha/2}$ is the $\alpha/2$ quantile of the distribution of $h(\theta, \hat{\theta}_n)$.

The qth quantile of the distribution of $h(\theta, \hat{\theta}_n)$ is approximated by the parametric bootstrap as

$$\tilde{v}_q = h(\hat{\theta}_n, \theta_n^*)_{[(M+1)q]}. \tag{10.9}$$

To obtain endpoints of an interval for θ then requires solving $h(L_\alpha, \hat{\theta}_n) = \tilde{v}_q$ for L and $h(U_\alpha, \hat{\theta}_n) = \tilde{v}'_q$ for U, where q and q' may be $\alpha/2$ or $1-\alpha/2$ depending on the situation.

Difference Discrepancy Function

Consider the discrepancy function $h(\theta, \hat{\theta}_n) = n^{1/2}(\hat{\theta}_n - \theta)$. This function reverses the inequalities in (10.6) so that

$$\begin{aligned}
\hat{\theta}_n - L_\alpha &\approx (\theta^*_{n,m} - \hat{\theta}_n)_{[(M+1)(1-\alpha/2)]} \\
\hat{\theta}_n - U_\alpha &\approx (\theta^*_{n,m} - \hat{\theta}_n)_{[(M+1)(\alpha/2)]}
\end{aligned}$$

which results in

$$\begin{aligned}
L_\alpha &\approx \hat{\theta}_n - (\theta^*_{n,m} - \hat{\theta}_n)_{[(M+1)(1-\alpha/2)]} \\
&= \hat{\theta}_n - (\theta^*_{n,m})_{[(M+1)(1-\alpha/2)]} + \hat{\theta}_n \\
&= 2\hat{\theta}_n - (\theta^*_{n,m})_{[(M+1)(1-\alpha/2)]},
\end{aligned}$$

and,

$$\begin{aligned}
U_\alpha &\approx \hat{\theta}_n - (\theta^*_{n,m} - \hat{\theta}_n)_{[(M+1)(\alpha/2)]} \\
&= \hat{\theta}_n - (\theta^*_{n,m})_{[(M+1)(\alpha/2)]} + \hat{\theta}_n \\
&= 2\hat{\theta}_n - (\theta^*_{n,m})_{[(M+1)(\alpha/2)]}.
\end{aligned}$$

A $(1-\alpha)100\%$ interval estimate for θ is then,

$$\left(2\hat{\theta}_n - (\theta^*_{n,m})_{[(M+1)(1-\alpha/2)]},\, 2\hat{\theta}_n - (\theta^*_{n,m})_{[(M+1)(\alpha/2)]}\right). \tag{10.10}$$

The interval (10.10) is called the *basic bootstrap confidence interval* for θ by Davison and Hinkley (1997), but it is really the basic interval that results from the discrepancy function $h(\theta, \hat{\theta}_n) = n^{1/2}(\hat{\theta}_n - \theta)$. Note that this interval estimate assumes that things have been arranged so that $(M+1)(\alpha/2)$ and $(M+1)(1-\alpha/2)$ are integers. This is not difficult if M is large; for example, take $\alpha/2 = 0.05$ and $M = 9999$ for a 90% interval.

The normal approximation interval given in expression (10.4) has the property of being symmetric, in this case about $\hat{\theta}_n - B(\hat{\theta}_n)$; it will be symmetric about $\hat{\theta}_n$ if that estimator is known to be unbiased, in which case we take $B(\hat{\theta}_n) = 0$ rather than estimating it. The interval (10.10) however, is not necessarily symmetric. It will be symmetric or close to symmetric as the empirical distribution of $\{\theta^*_{n,m} : m = 1, \ldots, M\}$ is symmetric or close to symmetric.

Ratio Discrepancy Function

Now consider using the discrepancy function $h(\theta, \hat{\theta}_n) = n^{1/2}(\theta/\hat{\theta}_n - 1)$. We would consider this function only for θ that are restricted to a strictly positive parameter space, in which case it does not reverse the inequality of (10.6) so that,

$$\begin{aligned}
\left[n^{1/2}\left(\frac{L_\alpha}{\hat{\theta}_n} - 1\right)\right] &\approx \left[n^{1/2}\left(\frac{\hat{\theta}_n}{\theta^*_n} - 1\right)\right]_{[(M+1)(\alpha/2)]} \\
\left[n^{1/2}\left(\frac{U_\alpha}{\hat{\theta}_n} - 1\right)\right] &\approx \left[n^{1/2}\left(\frac{\hat{\theta}_n}{\theta^*_n} - 1\right)\right]_{[(M+1)(1-\alpha/2)]}
\end{aligned}$$

These relations imply that

$$\begin{aligned}
\frac{L_\alpha}{\hat{\theta}_n} &\approx \left(\frac{\hat{\theta}_n}{\theta^*_n}\right)_{[(M+1)(\alpha/2)]} \\
\frac{U_\alpha}{\hat{\theta}_n} &\approx \left(\frac{\hat{\theta}_n}{\theta^*_n}\right)_{[(M+1)(1-\alpha/2)]}.
\end{aligned}$$

Now,

$$\begin{aligned}
\left(\frac{\hat{\theta}_n}{\theta^*_n}\right)_{[(M+1)(\alpha/2)]} &= \hat{\theta}_n \frac{1}{\theta^*_{n,[(M+1)(1-\alpha/2)]}} \\
\left(\frac{\hat{\theta}_n}{\theta^*_n}\right)_{[(M+1)(1-\alpha/2)]} &= \hat{\theta}_n \frac{1}{\theta^*_{n,[(M+1)(\alpha/2)]}}
\end{aligned}$$

resulting in

$$\begin{aligned} L_\alpha &\approx \hat{\theta}_n^2 \frac{1}{\theta^*_{n,[(M+1)(1-\alpha/2)]}} \\ U_\alpha &\approx \hat{\theta}_n^2 \frac{1}{\theta^*_{n,[(M+1)(\alpha/2)]}} \end{aligned}$$

and a $(1-\alpha)100\%$ interval estimate for θ is

$$\left(\hat{\theta}_n^2 \frac{1}{\theta^*_{n,[(M+1)(1-\alpha/2)]}},\ \hat{\theta}_n^2 \frac{1}{\theta^*_{n,[(M+1)(\alpha/2)]}} \right). \tag{10.11}$$

Studentized Discrepancy Function

As a final example, suppose we use the discrepancy function

$$h(\theta, \hat{\theta}_n) = \frac{\hat{\theta}_n - \theta}{[V(\hat{\theta}_n)]^{1/2}},$$

where $V(\hat{\theta}_n)$ denotes the estimated variance of $\hat{\theta}_n$. Notice that to make use of this disrepancy function in practice we need to be able to produce an estimated variance for $\hat{\theta}_n$ from any given sample. This function reverses the inequality of (10.6) so that,

$$\begin{aligned} \frac{\hat{\theta}_n - L_\alpha}{[V(\hat{\theta}_n]^{1/2})} &\approx \left(\frac{\theta_n^* - \hat{\theta}_n}{[V(\theta_n^*)]^{1/2}} \right)_{[(M+1)(1-\alpha/2)]} \\ \frac{\hat{\theta}_n - U_\alpha}{[V(\hat{\theta}_n]^{1/2})} &\approx \left(\frac{\theta_n^* - \hat{\theta}_n}{[V(\theta_n^*]^{1/2})} \right)_{[(M+1)(\alpha/2)]}. \end{aligned}$$

In this case,

$$\begin{aligned} L_\alpha &\approx \hat{\theta}_n - [V(\hat{\theta}_n)]^{1/2} \left(\frac{\theta_n^* - \hat{\theta}_n}{[V(\theta_n^*)]^{1/2}} \right)_{[(M+1)(1-\alpha/2)]} \\ U_\alpha &\approx \hat{\theta}_n - [V(\hat{\theta}_n)]^{1/2} \left(\frac{\theta_n^* - \hat{\theta}_n}{[V(\theta_n^*)]^{1/2}} \right)_{[(M+1)(\alpha/2)]} \end{aligned}$$

and a $(1-\alpha)100\%$ interval estimate for θ is

$$\left(\hat{\theta}_n - [V(\hat{\theta}_n)]^{1/2}\left(\frac{\theta_n^* - \hat{\theta}_n}{[V(\theta_n^*)]^{1/2}}\right)_{[(M+1)(1-\alpha/2)]}, \hat{\theta}_n - [V(\hat{\theta}_n)]^{1/2}\left(\frac{\theta_n^* - \hat{\theta}_n}{[V(\theta_n^*)]^{1/2}}\right)_{[(M+1)(\alpha/2)]}\right). \tag{10.12}$$

The interval in (10.12) is often called a *studentized* bootstrap confidence interval. Studentized intervals have been shown in simulation studies to have quite nice coverage properties, and are generally considered to be superior to basic intervals using a difference discrepancy function. The drawback to these intervals is that we must be able to compute an estimated variance for the estimator from the actual data and also each bootstrap data set. This is a drawback, of course, because the motivation for using a bootstrap procedure to obtain an interval estimate is decreased if we have other options available, which we almost certainly will if we can compute $V(\hat{\theta}_n)$ from either an exact or asymptotic result.

10.3 Percentile Bootstrap Intervals

We will briefly mention one other approach to the formulation of bootstrap confidence intervals, called *percentile methods* by Davison and Hinkley (1997, Chapter 5.3). The origin of the name is not intuitively obvious but presumably comes from the fact that we end up using the $(\alpha/2)$ percentile and $(1-\alpha/2)$ percentile of the empirical distribution of bootstrap estimates θ^* as interval endpoints, as will be shown below. Percentile methods have been modified (or adjusted) in a number of ways that seem to offer some improvement over normal approximation or basic bootstrap intervals in meeting nominal coverage goals, although assessments have been made primarily through the use of nonparametric bootstrap sampling (see, e.g., Davison and Hinkley

1997, Chapter 5.4 and Chapter 5.7). The performance of percentile methods in parametric bootstrap, which is our concern here, is less well understood.

Suppose that there exists some transformation of the estimator $\hat{\theta}_n$, say $\hat{\phi}_n \equiv W(\hat{\theta}_n)$, such that the distribution of $\hat{\phi}_n$ is known to be symmetric; for the moment the existence of such a function $W(\cdot)$ is all that matters, not its identity. Consider, then, applying this transformation to $\hat{\theta}_n$ and then using the basic bootstrap method with difference discrepancy function to form an interval for $\phi = W(\theta)$, with the following modifications. First, notice that we now have estimates of functionals of the distribution of $h(\phi, \hat{\phi}_n)$ through the bootstrap simulations $\{h(\hat{\phi}, \phi_n^*)_m : m = 1, \ldots, M\}$. In the basic bootstrap method we take $h(\phi, \hat{\phi}_n) = \hat{\phi}_n - \phi$ and $h(\hat{\phi}, \phi_n^*)_m = \phi_{n,m}^* - \hat{\phi}_n$. The development of the basic bootstrap interval for ϕ proceeds in the same way as before and we want $h(L_\alpha - \hat{\phi}_n) = \hat{\phi}_n - L_\alpha = v_{1-\alpha/2}$ and $h(U_\alpha - \hat{\phi}_n) = \hat{\phi}_n - U_\alpha = v_{\alpha/2}$ Now, however, the symmetry of the distribution of $\hat{\phi}_n - \phi$ indicates that $v_{1-\alpha/2} = -v_{\alpha/2}$ so that,

$$\begin{aligned}
\hat{\phi} - L_\alpha &= -v_{\alpha/2} \\
\hat{\phi} - U_\alpha &= -v_{1-\alpha/2},
\end{aligned}$$

where v_α is now a quantile from the distribution of $\hat{\phi}_n - \phi$. Then,

$$\begin{aligned}
L_\alpha &= \hat{\phi}_n + (\phi_n^* - \hat{\phi}_n)_{[(M+1)(\alpha/2)]} \\
&= (\phi_n^*)_{[(M+1)(\alpha/2)]} \\
U_\alpha &= \hat{\phi}_n + (\phi_n^* - \hat{\phi}_n)_{[(M+1)(1-\alpha/2)]} && (10.13) \\
&= (\phi_n^*)_{[(M+1)(1-\alpha/2)]}. && (10.14)
\end{aligned}$$

Now, if the transformation $W(\cdot)$ that produced ϕ from θ was monotone, then $\phi_{[k]}^*$ corresponds to $\theta_{[k]}^*$ for any integer $k \in \{1, \ldots, M\}$. Transforming

the interval endpoints (10.13) back to the θ scale then results in the bootstrap percentile interval for θ of,

$$\left((\theta_n^*)_{[(M+1)(\alpha/2)]},\ \ (\theta_n^*)_{[(M+1)(1-\alpha/2)]}\right). \tag{10.15}$$

What is surprising, then, is that such an interval can be formulated (and computed) for θ without ever determining what the transformation $\phi = W(\theta)$ might be.

10.4 Predication Intervals

Another use of parametric bootstrap is in forming prediction intervals for a new random variable Y^0, assumed to follow the same model as $Y_1, \ldots, Y_n$. Let the model evaluated at a possible value y^0 be denoted as $F(y^0|\theta)$. Given an estimated parameter $\hat{\theta}_n$ the natural starting point is an interval with endpoints given by the $(\alpha/2)$ and $(1-\alpha/2)$ quantiles of the estimated model $F(y^0|\hat{\theta})$. Denote these values as

$$\begin{aligned} q(\hat{\theta}_n)_{\alpha/2} &= F^{-1}(\alpha/2\,|\,\hat{\theta}_n) \\ q(\hat{\theta}_n)_{1-\alpha/2} &= F^{-1}(1-\alpha/2\,|\,\hat{\theta}_n). \end{aligned} \tag{10.16}$$

The interval $q_(\hat{\theta}_n)\alpha/2$, $q(\hat{\theta}_n)_{1-\alpha/2})$ will be overly optimistic (i.e., too short) because it does not take into account uncertainty in the estimation of θ by $\hat{\theta}_n$. That is, if we knew the true value θ, it would be the case that

$$Pr\left[q(\theta)_{\alpha/2} \leq Y^0 < q(\theta)_{1-\alpha/2}\,|\,\theta\right] = 1-\alpha.$$

Since we do not know θ but are estimating it with $\hat{\theta}_n$ we need to assess the actual coverage rate,

$$Pr\left[q(\hat{\theta}_n)_{\alpha/2} \leq Y^0 < q(\hat{\theta}_n)_{1-\alpha/2}\,|\,\theta\right] = 1-c(\alpha). \tag{10.17}$$

If there is a functional relation between $c(\alpha)$ and α, then we could "adjust" the procedure to use $q(\hat{\theta}_n)_{\alpha'/2}$ and $q(\hat{\theta}_n)_{1-\alpha'/2}$, where α' is chosen such that $c(\alpha') = \alpha$. The essential problem, then, is estimation of $c(\alpha)$ in expression (10.17). A parametric bootstrap may be used for this estimation in the following way.

The function of θ and $\hat{\theta}_n$ to be estimated is,

$$\begin{aligned} h(\theta, \hat{\theta}_n) &= Pr\left[q(\hat{\theta}_n)_{\alpha/2} \leq Y^0 < q(\hat{\theta}_n)_{1-\alpha/2} \,|\, \theta\right] \\ &= E\left\{I\left[q(\hat{\theta}_n)_{\alpha/2} \leq Y^0 < q(\hat{\theta}_n)_{1-\alpha/2} \,|\, \theta\right]\right\}. \end{aligned}$$

Given a fitted model through $\hat{\theta}_n$, simulate bootstrap data sets $\boldsymbol{y}^*_m \equiv (y^*_1, \ldots, y^*_n)^T_m$ in the usual way from $F(y_1, \ldots, y_n|\hat{\theta})$ to obtain bootstrap estimates $\theta^*_{n,m}$; $m = 1, \ldots, M$. Also simulate values of the predictand y^0_m; $m = 1, \ldots, M$ from the fitted model $F(y^0|\hat{\theta}_n)$. Compute the intervals $(q^*(\theta^*_n)_{(\alpha/2)},\ q^*(\theta^*_n)_{(1-\alpha/2)})_m$ with nominal coverage $1-\alpha$ for each bootstrap data set as in 10.16 with θ^*_n in place of $\hat{\theta}_n$.

Estimate the actual coverage of the interval as,

$$1 - c_M(\alpha) = \frac{1}{M}\sum_{m=1}^{M} I\left[\left(q^*(\theta^*_n)_{(\alpha/2)} \leq y^0_m < q^*(\theta^*_n)_{(1-\alpha/2)}\right)_m\right]. \tag{10.18}$$

Expression (10.18) is then a bootstrap estimate of the probability (10.17). There are then two options. One might simply report $1 - \hat{c}(\alpha)$ as the actual coverage, or one might relate $\hat{c}(\alpha)$ to α through some empirical model (e.g., a quadratic regression of $\hat{c}(\alpha)$ on α might provide a good description of the relation). In the latter case, we can attempt to select an appropriate value α' to use in expression (10.16) in calculating $q(\hat{\theta}_n)_{\alpha'/2}$ and $q(\hat{\theta}_n)_{1-\alpha'/2}$ to provide an actual coverage at level $1-\alpha$.

10.5 Dependence and Other Complications

The usefulness of parametric bootstrap is perhaps the greatest in situations for which we have an estimator $\hat{\theta}$ but it is difficult to derive the variance or distribution of $\hat{\theta}$. At the same time, we have presented parametric bootstrap methods for sets of independent random variables $Y_1, \ldots, Y_n$. This does seem somewhat incongruous, since it is situations in which we fail to have independence among response variables that most often leads to the inability to make use of distributional results for the purposes of inference. As pointed out by Davison and Hinkley (1997) the theoretical underpinnings of using bootstrap methods with models that contain complex dependence structures (e.g., spatial models) are both unresolved and an area of intensive research. This still remains true today, although any number of advances have been made since the late 1990s, largely in the area of nonparametric bootstrap. Nevertheless, the use of simulation from fitted parametric models seems to hold great potential for a large number of problems.

Consider problems which might be amenable to asymptotic inference. Underlying the development of theoretical properties of bootstrap estimators (either parametric or nonparametric) are two levels of asymptotics. At one level is the convergence of the distribution of $h(\hat{\theta}_n, \theta_m^*)$ computed from bootstrap data sets to the distribution of $h(\theta, \hat{\theta}_n)$. Here, we must recall that θ_m^* is a function of the bootstrap sample, $\boldsymbol{y}_m^* \equiv (y_1^*, \ldots, y_n^*)_m^T$, so that suppressed in this notation is the fact that each $\boldsymbol{y}_m^*$ is of dimension n and each θ_m^* is based on $\boldsymbol{y}_m^*$. The fact that bootstrap samples $\boldsymbol{y}_m^*$ have been independently generated certainly helps in demonstrating this convergence as M increases. A difficulty is if the distribution of $h(\theta, \hat{\theta}_n)$ depends heavily on the value of θ, since we are using $\hat{\theta}_n$ in the role of the true value of θ for bootstrap simula-

tions. The ideal situation is if $h(\theta, \hat{\theta}_n)$ is a *pivotal* quantity; recall this means that the distribution of $h(\theta, \hat{\theta}_n)$ is independent of θ. Unfortunately, we can often only demonstrate this in a definite manner for fairly simple models, in which case we may have alternative procedures than bootstrap for computing inferential quantities. We may, however, always examine the dependence of the distribution of $h(\theta, \hat{\theta}_n)$ on θ in the following way. Let $Q_M(h|\theta^{(k)})$ denote the estimated q^{th} quantile of $h(\theta, \hat{\theta}_n)$ based on a bootstrap simulation of size M with data generated from the model with parameter value $\theta^{(k)}$. That is, $Q_M(h|\theta^{(k)}) = h(\theta^{(k)}, \theta^*)_{[(M+1)q]}$. If values $Q_M(h|\theta^{(k)})$ are computed for a range of values, $\theta^{(k)} \in \{\hat{\theta}_n \pm k\delta : k = 1, \ldots, K\}$ for some δ, then a simple plot of $Q_M(h|\theta^{(k)})$ against $\theta^{(k)}$ may demonstrate the degree of dependence of $h(\theta, \hat{\theta}_n)$ on θ, or at least the relative dependence for several possible choices of $h(\cdot)$.

The second level of convergence needed arises because, in order for inference about θ based on $\hat{\theta}_n$ to have discernible properties, it is necessary that the distribution of $h(\theta, \hat{\theta}_n)$ allows some description of its probabilistic behavior as n grows large. This issue involves the proverbial *not loosing sight of the forest for the trees* as it applies to bootstrap methods, even outside of an asymptotic context. Consider, for example, estimating μ from a model that gives $Y_1, \ldots, Y_n \sim iid\, N(\mu, \sigma^2)$. My estimator of choice will be $\hat{\mu}_n = 0.1Y_{[2]} + 0.9Y_{[n-3]}$, where $Y_{[k]}$ denotes the kth largest value of the set $\{Y_i : i = 1, \ldots, n\}$. Note, among other things, that here $E\{\hat{\mu}_n\} = \mu$ so $\hat{\mu}_n$ is an unbiased estimator. By taking $h(\mu, \hat{\mu}_n) = (\hat{\mu}_n - \mu)$ I will be perfectly capable of estimating the distribution of $h(\mu, \hat{\mu}_n)$ through a parametric bootstrap, forming bootstrap intervals, and so forth. This clearly does not, however, offer any justification for my choice of $\hat{\mu}_n$ in the first place.

An additional issue that can sometimes arise in applications of parametric

bootstrap involves the combined issues of *simulation error* and *statistical error*. We present one hypothetical example to illustrate this.

A Multinomial Model

An important environmental characteristic of riverine ecosystems is the distribution of "sediment types" over the bottom of the river. Sediment types are often placed into fairly broad categories such as sand, silt, clay, and gravel. These categories have to do with a combination of particle size and organic matter content of the substrate. Sediment type is one of the factors that determine the abundances of many types of aquatic invertebrates and plants and, consequently, things that depend on them such as fish. Sediment type is also related to various characteristics of water quality, such as clarity and the availability of dissolved oxygen to aquatic life. Ecologists are interested in these relations both from the viewpoint of scientific understanding and to improve prediction of productivity. Consider small portion of a large river such as the Mississippi or Yangtze from which we have obtained n sediment samples from main channel, side channel and backwater habitats, each of which will be categorized as either clay, sand, silt, or gravel, depending on the predominant composition of the sample. Our primary objective is to compare the distributions of sediment types among the habitats. We might well approach this problem based on a multinomial data model with four categories. Suppose that we would like to use a parametric bootstrap to construct confidence intervals for the probabilities of sediment types. Maximum likelihood estimates of the multinomial parameters based on $n = 100$ samples were $\hat{\theta}_1 = 0.19$, $\hat{\theta}_2 = 0.33$, $\hat{\theta}_3 = 0.41$, and $\hat{\theta}_4 = 0.07$ for clay, sand, silt and gravel, respectively. In a parametric bootstrap, many simulated data

sets will result in 0 values for gravel, which makes estimation for these simulated realizations of the model difficult or impossible. It is not clear how this problem should be dealt with. One approach, which has been used in any number of similar situations, is to condition the simulation estimates of uncertainty on estimable data sets by simply discarding any simulated data sets that do not allow estimation. This must result in an underestimation of uncertainty, but uncertainty about *what*; the values of parameters, the model itself, or the error of simulation? Put another way, should the simulation of unestimable data sets from an otherwise perfectly acceptable model impact error of simulation, statistical error in estimation, or error in model selection?